

EXPERIMENT – 4

AIM: To perform and analyze SQL queries using INNER JOIN, LEFT JOIN, SELF JOIN, 3way JOIN, correlated subqueries, EXISTS, and simulated INTERSECT and EXCEPT operations on the Employee–Department–Project database, and to compare their execution plans using EXPLAIN.

SOURCE CODE: USING EXPERIMENT – 3 DATABASE

SQL QUERIES:

INNER JOIN

SELECT

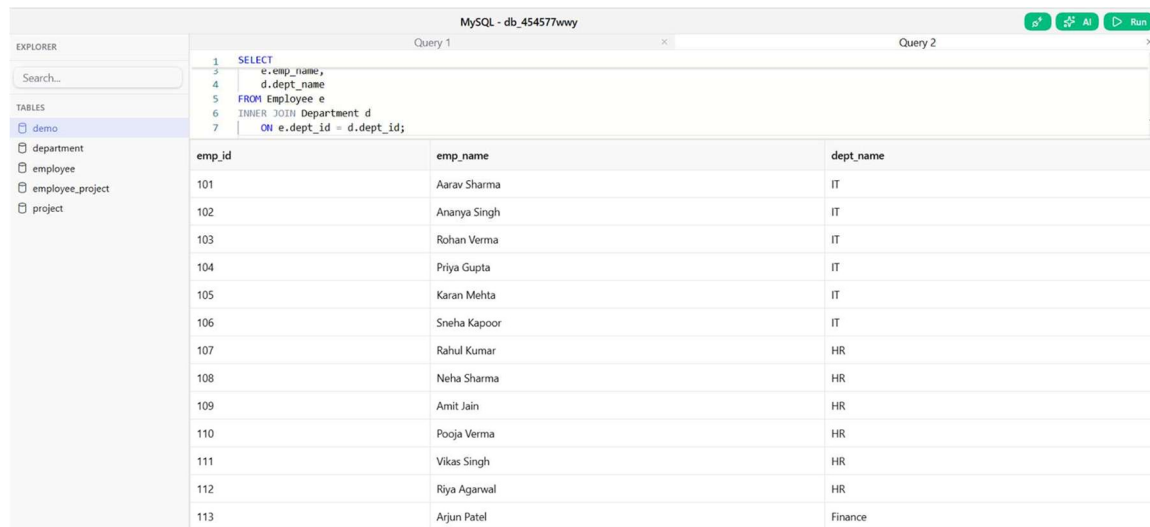
e.emp_id,
e.emp_name,
d.dept_name

FROM Employee e

INNER JOIN Department d

ON e.dept_id = d.dept_id;

OUTPUT:



The screenshot shows a MySQL database interface with a table explorer on the left and a query editor on the right. The table explorer lists the following tables: department, employee, employee_project, and project. The query editor displays the following SQL query:

```
1 SELECT  
2   e.emp_name,  
3   d.dept_name  
4 FROM Employee e  
5 INNER JOIN Department d  
6   ON e.dept_id = d.dept_id;
```

The results of the query are displayed in a table with the following columns: emp_id, emp_name, and dept_name. The results are as follows:

emp_id	emp_name	dept_name
101	Aarav Sharma	IT
102	Ananya Singh	IT
103	Rohan Verma	IT
104	Priya Gupta	IT
105	Karan Mehta	IT
106	Sneha Kapoor	IT
107	Rahul Kumar	HR
108	Neha Sharma	HR
109	Amit Jain	HR
110	Pooja Verma	HR
111	Vikas Singh	HR
112	Riya Agarwal	HR
113	Arjun Patel	Finance

LEFT JOIN

SELECT

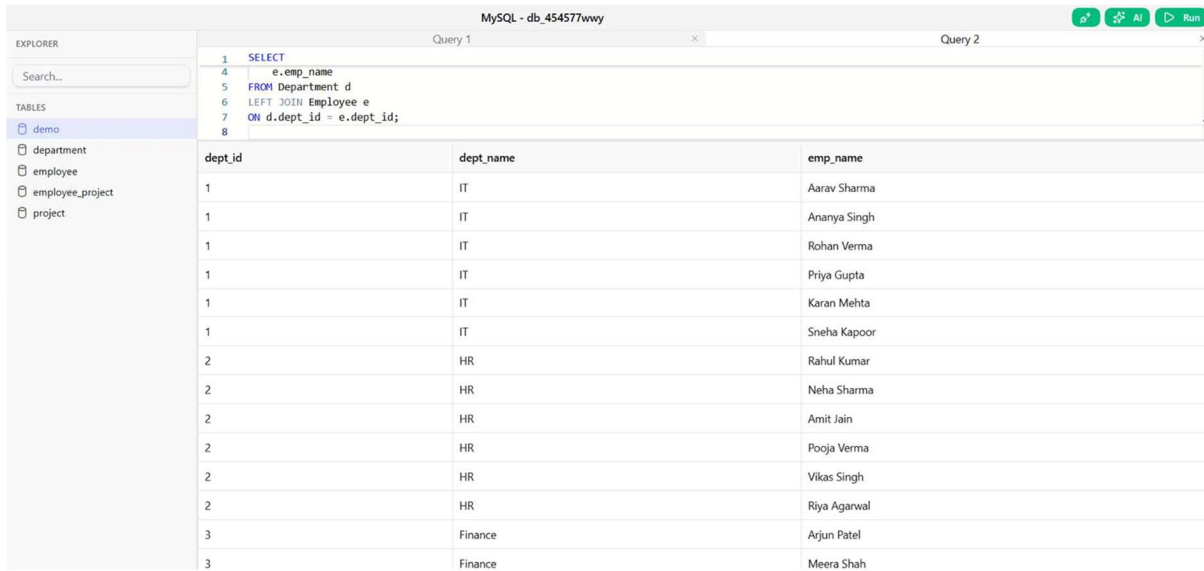
d.dept_id,
d.dept_name,
e.emp_name

FROM Department d

LEFT JOIN Employee e

ON d.dept_id = e.dept_id;

OUTPUT:



The screenshot shows a MySQL IDE window titled 'MySQL - db_454577wwy'. The 'EXPLORER' panel on the left shows a 'demo' database with tables: department, employee, employee_project, and project. The 'Query 1' editor contains the following SQL query:

```
1 SELECT
4   e.emp_name
5 FROM Department d
6 LEFT JOIN Employee e
7 ON d.dept_id = e.dept_id;
8
```

The output of the query is displayed in a table with the following data:

dept_id	dept_name	emp_name
1	IT	Aarav Sharma
1	IT	Ananya Singh
1	IT	Rohan Verma
1	IT	Priya Gupta
1	IT	Karan Mehta
1	IT	Sneha Kapoor
2	HR	Rahul Kumar
2	HR	Neha Sharma
2	HR	Amit Jain
2	HR	Pooja Verma
2	HR	Vikas Singh
2	HR	Riya Agarwal
3	Finance	Arjun Patel
3	Finance	Meera Shah

3-WAY JOIN

SELECT

e.emp_name,
d.dept_name,
p.project_name

FROM Employee e

JOIN Department d

ON e.dept_id = d.dept_id

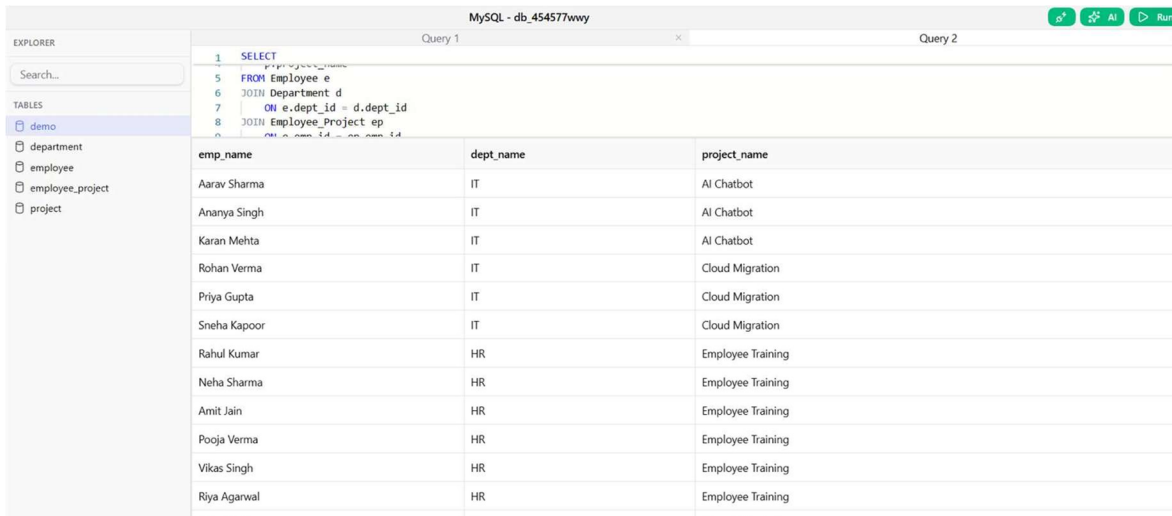
JOIN Employee_Project ep

ON e.emp_id = ep.emp_id

JOIN Project p

ON ep.project_id = p.project_id;

OUTPUT:



The screenshot shows a MySQL IDE window titled 'MySQL - db_454577wwy'. The 'EXPLORER' panel on the left shows a 'demo' database with tables: department, employee, employee_project, and project. The 'Query 1' editor contains the following SQL query:

```
1 SELECT
5   e.emp_name,
6   d.dept_name,
7   p.project_name
8 FROM Employee e
9 JOIN Department d
10 ON e.dept_id = d.dept_id
11 JOIN Employee_Project ep
12 ON e.emp_id = ep.emp_id
13 JOIN Project p
14 ON ep.project_id = p.project_id;
```

The output of the query is displayed in a table with the following data:

emp_name	dept_name	project_name
Aarav Sharma	IT	AI Chatbot
Ananya Singh	IT	AI Chatbot
Karan Mehta	IT	AI Chatbot
Rohan Verma	IT	Cloud Migration
Priya Gupta	IT	Cloud Migration
Sneha Kapoor	IT	Cloud Migration
Rahul Kumar	HR	Employee Training
Neha Sharma	HR	Employee Training
Amit Jain	HR	Employee Training
Pooja Verma	HR	Employee Training
Vikas Singh	HR	Employee Training
Riya Agarwal	HR	Employee Training

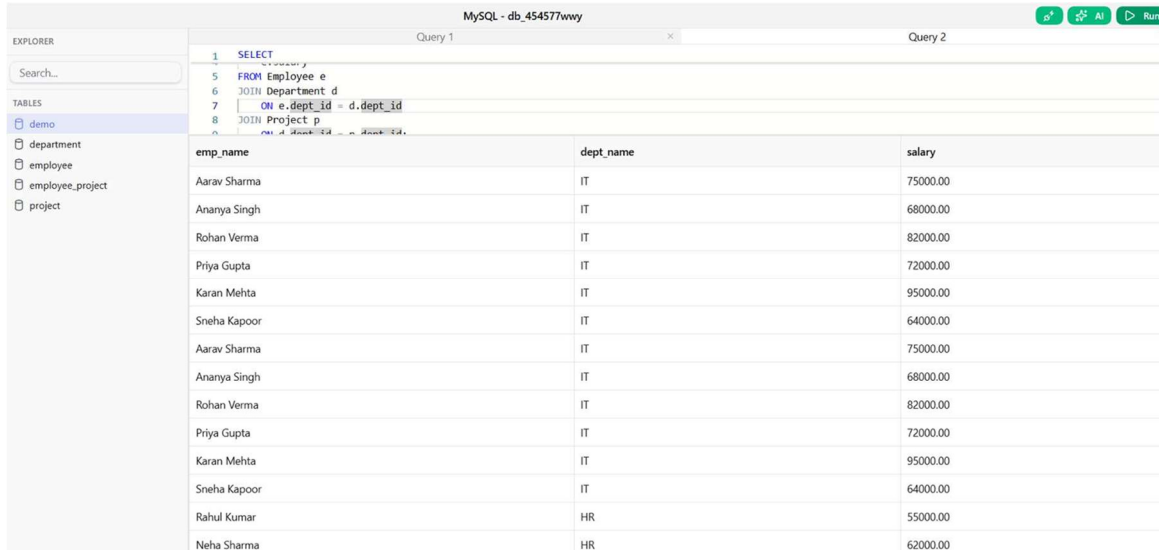
SELECT

e.emp_name,

```

        d.dept_name,
        e.salary
FROM Employee e
JOIN Department d
        ON e.dept_id = d.dept_id
JOIN Project p
        ON d.dept_id = p.dept_id;
OUTPUT:

```



The screenshot shows a MySQL Workbench interface with a query window titled 'Query 2'. The query is as follows:

```

1 SELECT
2     e.emp_name,
3     d.dept_name,
4     e.salary
5 FROM Employee e
6 JOIN Department d
7     ON e.dept_id = d.dept_id
8 JOIN Project p
9     ON d.dept_id = p.dept_id;

```

The output of the query is displayed in a table with the following data:

emp_name	dept_name	salary
Aarav Sharma	IT	75000.00
Ananya Singh	IT	68000.00
Rohan Verma	IT	82000.00
Priya Gupta	IT	72000.00
Karan Mehta	IT	95000.00
Sneha Kapoor	IT	64000.00
Aarav Sharma	IT	75000.00
Ananya Singh	IT	68000.00
Rohan Verma	IT	82000.00
Priya Gupta	IT	72000.00
Karan Mehta	IT	95000.00
Sneha Kapoor	IT	64000.00
Rahul Kumar	HR	55000.00
Neha Sharma	HR	62000.00

CORRELATED SUBQUERY

```

SELECT
    e.emp_id, e.emp_name, e.salary, e.dept_id
FROM Employee e
WHERE e.salary >
(
    SELECT AVG(e2.salary)
    FROM Employee e2
    WHERE e2.dept_id = e.dept_id
);

```

OUTPUT:

MySQL - db_454577www				
Query 1				
Query 2				
<pre> 1 SELECT 2 emp_id, 3 emp_name, 4 salary, 5 dept_id 6 FROM Employee e 7 WHERE e.salary > 8 (9 SELECT AVG(e2.salary) 10 FROM Employee e2 11) </pre>				
emp_id	emp_name	salary	dept_id	
103	Rohan Verma	82000.00	1	
105	Karan Mehta	95000.00	1	
110	Pooja Verma	67000.00	2	
112	Riya Agarwal	71000.00	2	
113	Arjun Patel	88000.00	3	
114	Meera Shah	92000.00	3	
116	Kavya Iyer	85000.00	3	
118	Nisha Reddy	90000.00	3	
121	Mohit Bansal	74000.00	4	
122	Isha Malhotra	81000.00	4	
124	Tanya Arora	77000.00	4	
125	Dev Malhotra	98000.00	5	
126	Aditi Mishra	91000.00	5	
128	Shreya Das	93000.00	5	

EXISTS

SELECT

```

    e.emp_id,
    e.emp_name
FROM Employee e
WHERE EXISTS
(
    SELECT 1
    FROM Employee_Project ep
    WHERE ep.emp_id = e.emp_id
);

```

OUTPUT:

MySQL - db_454577www				
Query 1				
Query 2				
<pre> 4 FROM Employee e 5 WHERE EXISTS 6 (7 SELECT 1 8 FROM Employee_Project ep 9 WHERE ep.emp_id = e.emp_id 10) </pre>				
emp_id	emp_name			
101	Aarav Sharma			
102	Ananya Singh			
103	Rohan Verma			
104	Priya Gupta			
105	Karan Mehta			
106	Sneha Kapoor			
107	Rahul Kumar			
108	Neha Sharma			
109	Amit Jain			
110	Pooja Verma			
111	Vikas Singh			
112	Riya Agarwal			
113	Arjun Patel			

SIMULATED INTERSECT

SELECT

```

    e.emp_id,
    e.emp_name
FROM Employee e

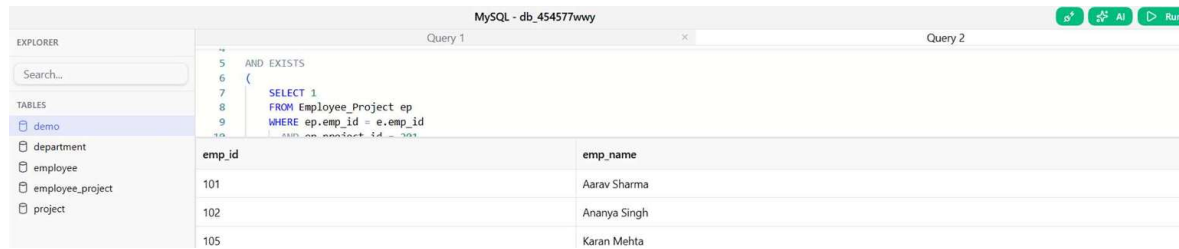
```

```

WHERE e.dept_id = 1
AND EXISTS
(
    SELECT 1
    FROM Employee_Project ep
    WHERE ep.emp_id = e.emp_id
        AND ep.project_id = 201
);

```

OUTPUT:



emp_id	emp_name
101	Aarav Sharma
102	Ananya Singh
105	Karan Mehta

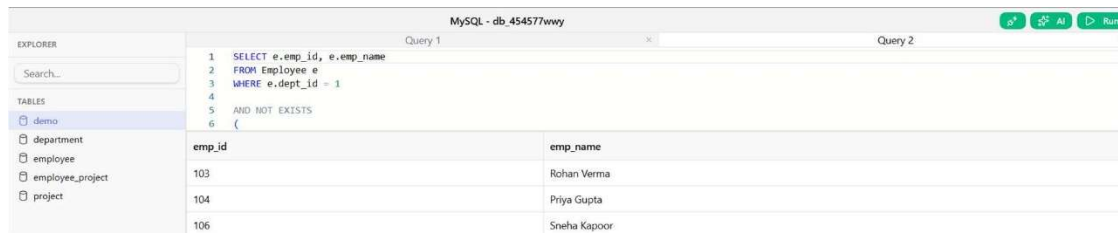
SIMULATED EXCEPT

```

SELECT e.emp_id, e.emp_name FROM Employee e
WHERE e.dept_id = 1
AND NOT EXISTS (
    SELECT 1
    FROM Employee_Project ep WHERE ep.emp_id = e.emp_id
    AND ep.project_id = 201
);

```

OUTPUT:



emp_id	emp_name
103	Rohan Verma
104	Priya Gupta
106	Sneha Kapoor

COMPARING EXECUTION PLANS USING EXPLAIN EXPLAIN – INNER JOIN

EXPLAIN

```

SELECT
e.emp_id,
e.emp_name,
d.dept_name
FROM Employee e
INNER JOIN Department d
ON e.dept_id = d.dept_id;

```

OUTPUT:

MySQL - db_454577www											
Query 1											
1	EXPLAIN										
2	SELECT										
3		e.emp_id,									
4		e.emp_name,									
5		d.dept_name									
6	FROM	Employee e									
id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	null	ALL	PRIMARY	null	null	null	5	100	null
1	SIMPLE	e	null	ref	dept_id	dept_id	5	db_454577www.d.dept_id	6	100	null

EXPLAIN – LEFT JOIN

EXPLAIN

SELECT

d.dept_name, e.emp_name

FROM Department d LEFT JOIN Employee e

ON d.dept_id = e.dept_id;

OUTPUT:

MySQL - db_454577www											
Query 1											
1	EXPLAIN										
2	SELECT										
3		d.dept_name,									
4		e.emp_name									
5	FROM	Department d									
6	LEFT JOIN	Employee e									
id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	null	ALL	null	null	null	null	5	100	null
1	SIMPLE	e	null	ref	dept_id	dept_id	5	db_454577www.d.dept_id	6	100	null

EXPLAIN – CORRELATED SUBQUERY

EXPLAIN SELECT

```
e.emp_name, e.salary
FROM Employee e
WHERE e.salary >
(
    SELECT AVG(e2.salary)
    FROM Employee e2
    WHERE e2.dept_id = e.dept_id
)
```

OUTPUT:

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	PRIMARY	e	null	ALL	null	null	null	null	30	100	Using where
2	DEPENDENT SUBQUERY	e2	null	ref	dept_id	dept_id	5	db_454577wwy.e.dept_id	6	100	null

EXPLAIN – EXISTS

EXPLAIN SELECT

```
e.emp_id, e.emp_name
FROM Employee e
WHERE EXISTS
(
    SELECT 1
    FROM Employee_Project ep
    WHERE ep.emp_id = e.emp_id
);
```

OUTPUT:

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	e	null	ALL	PRIMARY	null	null	null	30	100	null
1	SIMPLE	<subquery2>	null	eq_ref	<auto_distinct_key>	<auto_distinct_key>	4	db_454577wwy.e.emp_id	1	100	null
2	MATERIALIZED	ep	null	index	PRIMARY	project_id	4	null	30	100	Using index